

4.1/4.2 Continuous Random Variables

A continuous random variable X is a random variable whose range has uncountably infinite values. So the range of a continuous random variable must be an interval (either finite or infinite) of real numbers.

A continuous random variable is defined by a probability density function $f(x)$. This is a function such that

- (1) $f(x) \geq 0$ ($\forall x$)
- (2) $\int_{-\infty}^{\infty} f(x) dx = 1$ “all probabilities add up to 1”
- (3) $\int_c^d f(x) dx = P(c < X < d)$

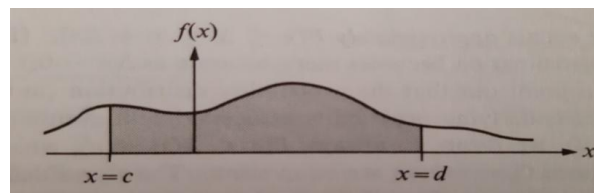


Figure 4.1 Probability determined from the area under $f(x)$

Remark

Because the magnitude of an area is not influenced by whether or not the edge of an area is counted, the following applies to continuous random variables:

$$\begin{aligned}
 P(c < X < d) &= P(c \leq X \leq d) \\
 &= P(c \leq X < d) \\
 &= P(c < X \leq d)
 \end{aligned}$$

Example 1

A shooter tries to hit a target. The distance from the point of impact to the bull's-eye is a continuous random variable X with probability density function:

$$f(x) = 1 - \frac{1}{2}x \quad \text{for } 0 \leq x \leq 2$$

There is a hit if the distance is no more than $\frac{1}{2}$. Determine $P(\text{hit})$.

Solution

See Figure 4.2.

$$\begin{aligned}
 P(\text{hit}) &= P(X \leq 0.5) \\
 &= \int_0^{0.5} \left(1 - \frac{1}{2}x\right) dx \\
 &= \left[x - \frac{1}{4}x^2\right]_{x=0}^{x=0.5} \\
 &= \frac{1}{2} - \frac{1}{4} \cdot \left(\frac{1}{2}\right)^2 = 0.4375
 \end{aligned}$$

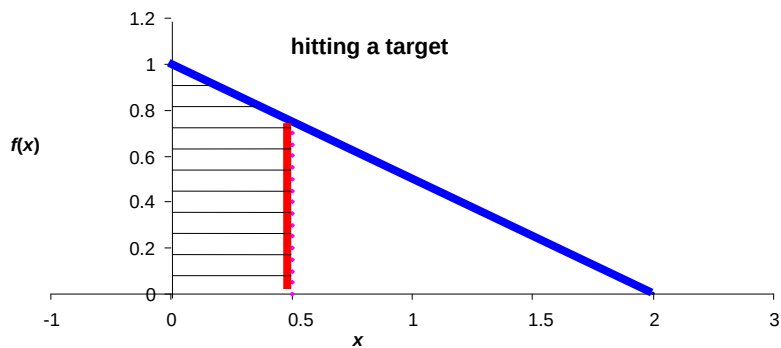


Figure 4.2 Probability density function for Example 1

Remark

Using `gc/math/option 9`: $\int_0^{0.5} 1 - 0.5x \, dx = 0.4375$

4.3 Cumulative Distribution Functions

From $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) \, dt$ it follows immediately:

$$f(x) = \frac{d}{dx} F(x)$$

Example 2

(see the data in Example 1)

$$F(x) = \begin{cases} 0 & , \quad \text{for } x < 0 \\ \int_0^x \left(1 - \frac{1}{2}t\right) dt = \left[t - \frac{1}{4}t^2\right]_0^x = x - \frac{1}{4}x^2 & , \quad \text{for } 0 \leq x \leq 2 \\ 1 & , \quad \text{for } x > 2 \end{cases}$$

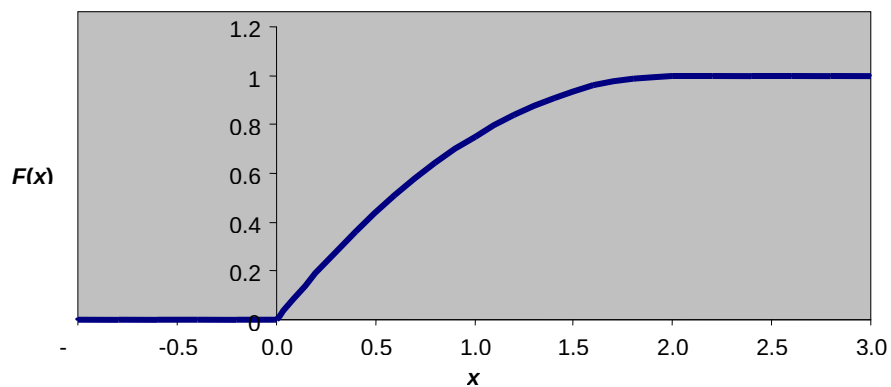


Figure 4.3 Cumulative distribution function for Example 2

4.4 Mean and Variance of a Continuous Random Variable

Analogous to the definition of the discrete case, the following applies to continuous random variables

$$\mu = E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$\sigma^2 = V(X) = \begin{cases} E(X - \mu)^2 & = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f(x) dx \\ E(X^2) - E(X)^2 & = \int_{-\infty}^{\infty} x^2 \cdot f(x) dx - \mu^2 \end{cases}$$

Example 3

(see the data in Example 1)

$$\begin{aligned} \bullet \quad E(X) &= \int_0^2 x \cdot \left(1 - \frac{1}{2}x\right) dx = \int_0^2 \left(x - \frac{1}{2}x^2\right) dx \\ &= \left[\frac{1}{2}x^2 - \frac{1}{6}x^3\right]_0^2 = 2 - 4/3 = 2/3 \\ \bullet \quad V(X) &= E(X^2) - E(X)^2 = \int_0^2 x^2 \cdot \left(1 - \frac{1}{2}x\right) dx - \left(\frac{2}{3}\right)^2 \\ &= \left[\frac{1}{3}x^3 - \frac{1}{8}x^4\right]_0^2 - \frac{4}{9} = 8/3 - 2 - 4/9 = 2/9 \end{aligned}$$

4.5 Continuous Uniform Distribution

A real number is generated randomly in the interval $[a; b]$. All real numbers between a and b are therefore equally likely.

So

$$f(x) = \text{constant}$$

The continuous random variable X is defined as follows:

X = the size of the selected real number

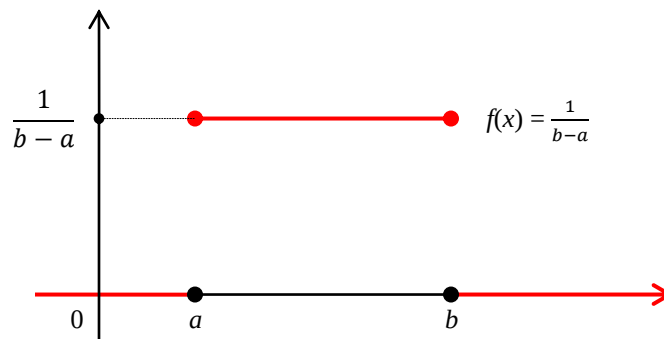


Figure 4.4

Continuous uniform probability density function

The random variable X has a continuous uniform distribution over the interval $[a; b]$.

Remarks

- Notation: $X \sim u[a; b]$
- If $a \leq c < d \leq b$, then

$$P(c < X < d) = \int_c^d \frac{1}{b-a} dx = \left[\frac{x}{b-a} \right]_c^d = \frac{d-c}{b-a}$$

So the magnitude of the probability is determined solely by the ratio between the **length of [c; d]** and the **length of [a; b]**.

e.g. if $X \sim u[1; 7]$, then

$$\begin{aligned} P(2 \leq X < 4) &= \frac{4-2}{7-1} = 0.3333 \\ P(X \geq 6) &= \frac{7-6}{7-1} = 0.1667 \end{aligned}$$

- $\mu = E(X) = \frac{1}{2}(a+b)$ the middle of $[a; b]$
- $\sigma^2 = V(X) = \frac{1}{12}(b-a)^2$

Proof (only the formula for the expected value)

$$\begin{aligned} E(X) &= \int_a^b x \cdot \frac{1}{b-a} dx = \left[\frac{1}{b-a} \cdot \frac{1}{2} x^2 \right]_a^b = \frac{\frac{1}{2}(b^2-a^2)}{b-a} \\ &= \frac{\frac{1}{2}(b+a) \cdot (b-a)}{b-a} = \frac{1}{2}(a+b) \end{aligned}$$

Example 4

Suppose $X \sim u[1; 7]$ and $Y = -3X + 2$.

Determine $E(X)$, $E(Y)$, $V(X)$ and $V(Y)$.

Solution

$$\begin{aligned} E(X) &= \frac{1}{2}(1+7) = 4 \\ V(X) &= \frac{1}{12}(7-1)^2 = 3 \\ E(Y) &= -3 \cdot E(X) + 2 = -3 \cdot 4 + 2 = -10 \\ V(Y) &= (-3)^2 \cdot V(X) = 9 \cdot 3 = 27 \end{aligned}$$